

# Chapter 14

## “Bildung” as a Forgotten Aspect of Algorithmic Technologies



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### 14.1 Introduction

As part of the “Global Educational Industry” (GEI), developers, coders, and designers of digital technologies such as hardware, software and learning platforms have a significant influence on the understanding of education and learning (Parreira do Amaral et al., 2019). GEI is described as “central to the transformation of education from a modern to a late modern institution” (Amos, 2019, p. 226). This concerns “the capitalization of the educational sector on a global scale” (Thompson & Parreira do Amaral, 2019, p. 3), mobilized partly by the so-called disruptive innovation of digital technologies and data infrastructures.

By introducing “proprietary” digital technologies and data science to formal and informal education, technology enterprises concentrate “the means for the production of knowledge in their own hands” (Williamson, 2017, p. 118). Dieter et al. (2019) call this predominance of certain businesses “Silicon Valley imperialism” (p. 12). Digital technologies, such as learning analytics (Allert et al., 2018), apps (Decuypere, 2019) or learning software, are not pedagogically neutral but hold assumptions about the subject and they stage learning in a certain way: through *algorithmic* information processing, knowledge construction and educational practices are visualized in particular ways.

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Our contribution focuses on this kind of educational framing by learning software for schools and its implications for understanding teaching and learning practices. From this perspective, the fundamental concept of education as “Bildung” comes into sight. We explore this theoretically by discussing first how sense-making within algorithmic systems can be described; second, we explore how meaning is staged for and with the user through visualization and gamification; and third, we discuss terms used in the theory of education and specifically “Bildung”. Against that background, we will draw on specific examples of learning software used in the school context, focusing the way knowledge is assessed, how the reward system is constructed and how performance is visualized. Finally, we discuss aspects of the algorithmic framing of pedagogy in terms of their lack of opportunities for sense-making and “Bildung” and what this means for human sense-making in general in the field of education.

## 14.2 Algorithmic Sense-Making in the Context of Knowledge Construction

Digital technologies are data-based, which means that they *need* data and a data infrastructure to process and produce information. This applies to all programs regardless of their complexity, from simple algorithmic software programs to highly complex applications for machine learning. In that sense, only sequenced, abstract – digital – data is of use in computer technology, and in consequence analog information has to be digitized, for example via analog-to-digital converters or by prompting users to enter information manually (Berry, 2011). Within digital computing technology, programs are built around algorithms that define how data is collected and processed and how the results are presented. Accordingly, algorithms are a guide to how any input can be transformed into a desired output (Stalder, 2016, p. 167).

Therefore, data is neither “just there” nor does it mean anything by itself. Digital data are the result of translation processes of the analog world, while algorithms – as codes – are mechanisms to handle the data: “As such, code is operative and produces a result [...] often in an iterative process of loops and conditionals” (Berry, 2011, p. 52). Even self-learning systems (e.g. artificial intelligence, machine learning) need to be told how and what to learn: Algorithms are needed to access, select, process, interpret and present data for particular purposes.

To interlink “the analog” and “the digital”, certain ways of structuring and classifying information are necessary. Since the 1970s computer technology development and design practice have worked on designing technology that is user-friendly, ergonomic, convenient and seamless by researching and configuring human behavior itself (Suchman, 1985; Wiener, 1950; Woolgar, 1990). Hence, design and technical functionality means that human-computer interaction has become almost seamlessly integrated into our daily practices.

We want to take one step back, and point out the conditions that enable this intertwining of human and digital technological practices. Humans become detectable for digital technologies via stimulus-response relationships and their behavior, whereby non-observable dimensions, such as thinking and experiencing or feeling, have to be (momentarily) neglected. It must be noted that non-observable dimensions such as feelings or well-being are also made computable or machine-readable, for example, via face tracking; so it has to be transferred into a dimension, which is traceable. Despite these sophisticated tracking practices and the complex information produced by big data computation and machine learning, all algorithmic programs follow unambiguous schematic operational steps. All those data processing practices refer to objects that, via digitization, are independent of the materiality of the analog objects (Krämer, 1988, pp. 159–160). Hence, an individual step has to be measured via sensors and collected and transformed within computer technology into binary data in the same way as student answers to a test. Within algorithmic programs, all binary data are handled as the same kind of information. Different content or its mediality, such as images, text, and sound, are transported, processed, and stored as data in the same way (Manovich, 2013, p. 133). Therefore, information is rendered as a non-specific “thing” in order to have the same meaning in every context (Buckland, 1991).

Genealogically speaking, understanding information in this rationalistic sense, and the receding of a didactic sense, can be dated to the rise of information theory in the industrialized world. It intensified after the Second World War with cybernetics as a theory of communication with the idea of controlling man, machine and processes. Since then, information has had to be transferable between humans and machines as much as between machines only. This is why information needs to be “comprehensible” on both sides (Kübler, 2009). Information therefore has to be, so to speak, ‘meaningless’, for an information transfer model that addresses humans and machines in the same way. The cybernetic model went so far as to construct learning for humans and machines in a circuit system via *regulation* of information: Data is received, stored and compared (with former data), chosen and externalized (Wiener, 1972, p. 114). One way of pre-sorting the world and the human being, in order to be able to predict and interpret human behavior, is to make it measurable through quantification. This creation of measurable constructs through quantification appears to be a necessary condition for the already discussed fragmentation and digitization of knowledge to produce supposedly unambiguous content. It also creates certain conditions for dealing with knowledge; among other things, it makes it possible to render action quantifiable by technology. There are two sides of quantification in and through software: one related to input and one related to output. On the input side, data is quantified via sensor-technology, data collection, data abstraction, and data adjustment – here, it is necessary to decide which data has to be digitized in what way. On the output side the interpreted data has to be staged and communicated as quantified and/or visualized and verbalized in relation to the user. Quantified outputs (such as ‘You walked 3,700 steps today’, ‘You solved 80% of the assignment’) are based on technical functions mentioned above, but also convey a certain kind of value. Both input and output algorithmic data processing can be

described partly as a practice of the quantitative interpretation of the world in order to grasp and understand it through software.

Quantification as a *social* technology has a long tradition that “pushes the value of accuracy forward” and “increases confidence in the reliability, objectivity and safety of actions and the objects and technical devices they produce” (Hörning, 2001, p. 126, translated by DK). Although quantification suggests objectivity, comparability is not inherent in the objects but rather is a product of many decisions and standardization efforts (Heintz, 2010, p. 169). In the field of pedagogy, for example, the practices of quantification, comparison and measuring produce and change simultaneously what is measured (Manhart, 2016). For instance, school grades always imply pedagogical feedback. Poor grades could encourage learners to do better and learn more, discourage unwanted behavior or frustrate students. Conversely, good marks could be received as praise and recognition of hard work or they could invite students to sit back and relax. Either way, pedagogical measurement and quantification always has an impact.

To produce output and, therefore, information to keep the algorithmic system going, algorithms work on two layers: while the data processing code-layer (Application Program Interface or API) is invisible to users, some results of the processing are made visible and manipulable on the graphical user interface (GUI). On that visible layer, the users have to feel they are being referred to and can make sense of what is offered.

The construction of *meaning* also takes place in algorithmic systems on different levels. First, the social-discursive knowledge of the corresponding developers’ milieu is inscribed (Klinge, 2018). Since the programming of algorithms and the design of computer technology are “modelled on visions of the social world” (Beer, 2016, p. 4), they are always documents of anthropological understandings and reflect contemporary values (Seaver, 2018). Developing digital technologies entails a process of constructing the social, because “by setting parameters for the user’s actions, the evolving machine effectively attempts to configure the user” (Woolgar, 1990, p. 61). Second, however, the user creates meaning on the basis of information produced by the digital technology in various contexts, coming to understand aspects of the world *through* digital systems. Thus, it can be argued that algorithmic systems also change human ways of constructing meaning.

Both levels of meaning construction are mediated by algorithmic functionality (Klinge, 2020). Instant feedback on human actions is needed to maintain the users’ interaction with the program, but also its reliability and continuity seems to act as a “buffer zone against the uncertainties and worries of their world” (Schüll, 2014, p. 13). Rouvroy (2013) describes this way of meaning construction as “data behaviorism”:

The ‘real time operability’ of devices functioning on such algorithmic logic saves human actors the burden and responsibility to transcribe, interpret and evaluate the events of the world. It saves them the meaning-making processes of transcription or representation, institutionalization, convention and symbolization (p. 143).

In this kind of knowledge there is no linearity of time from past into future, just synchrony (Esposito, 2013, p. 132).

Knowledge thus appears to be always available and retrievable, but not in the sense of a necessary acquisition in terms of learning or “Bildung”. In contrast to educational theories about cognition and acquisition, it can be said that the algorithmic construction of knowledge seems diametrically opposed to the concept of knowledge in education and educational sciences, which describes knowledge as a subjective construction of meaning and the acquisition of content, abilities and skills, which necessarily takes time. Communication, with its different layers of meaning, seems then to carry an essential dimension of knowledge, which includes reflection, criticism, responsibility, and ethics. From an epistemic point of view, digital technologies and their algorithmic program both inscribe and produce a certain kind of knowledge. They produce certain spheres of experience. From an educational science point of view, this raises the question how digital data as bits of sequenced information can be transferred into complex knowledge, and how learning is staged. From this perspective, we will see later how the feedback tools and reward systems of learning platforms are a quantified digital version of physical social interactions between the teacher and students.

### 14.3 Visualization and Gamification to Provoke Interaction

To maintain the algorithmic functionality, the technology needs data to process. Through the Graphical User Interface (GUI) visible affordances of the system can engage the user to interact with the system and gain data, which is then processed by the hidden affordances of the API or ‘black box’, invisible to the user (Berry, 2011, pp. 15–16). This division of code and interface was not present in early software development, when users had to operate the system via graphical elements as well as via code writing (Manovich, 2013, p. 98). The operation of computer technologies has changed fundamentally: most users are unable, and have no need, to control anything via code, and visual design has become the essential intersection of human-computer interaction. Interaction is possible by means of touch, swipe and prompt to enter information through the graphical user interface. This human-computer interaction involves signs, gestures, writing and reading, speaking and listening, showing and following, touching and triggering in symbolic framings where iconography directs the activities (Rammert, 2010, p. 49). To call someone via smartphone, you only need to touch the icon of a telephone and choose a name from your list. If you want to listen to music, you can tell Amazon’s Alexa, Apple’s Siri or Google’s Assistant which song you want to listen to. If you are learning Spanish with an app, you touch pictures to answer the questions, for example. The design compatibility for smaller surfaces such as mobile phones demands a more minimalist layout on the one hand, and addresses the users as “playfellows” in a colorful trivialized version on the other hand (Bunz, 2015, p. 194).

Many of the elements intended to provoke and maintain users' interactions are designed according to the principles of "gamification" (Zichermann & Cunningham, 2011). According to the technology industry, game elements should motivate users and make use of technology fun but also regulate human behavior:

To further engage our audiences, we need to consider reward structures, positive reinforcement, and subtle feedback loops alongside mechanisms such as scores, badges, levels, challenges, and leaderboards. When done well, gamification helps align our interests with the intrinsic motivations of our players, amplified with the mechanics and rewards that make them come in, bring friends, and keep coming back. Only by carefully unpacking consumer emotions and desires can we design something that really sticks – and only through the power of gamification can we make that experience predictable, repeatable, and financially rewarding (Zichermann & Cunningham, 2011, pp. ix–x).

This quote from one of the more influential books on gamification in web and app design shows quite clearly the spirit of hope for designers and overall tech-industries that gamification can open up a new gold mine by subtly but significantly influencing and predicting human behavior. Feedback design uses badges, stars, points, diamonds, or abstract tokens to reward, motivate, and take the user to a desirable goal *inherent in the program*. The introduction of game elements to non-game contexts, such as education, organizations or the home environment by social media entrepreneurs and marketing experts, puts into practice the aim to efficiently regulate behavior via positive feedback (Fuchs et al., 2014).

The introduction of this principle to educational contexts bears significant problems for learning processes. Put simply, learning by points, stars and badges doesn't support an understanding of content and acquisition of deep knowledge, but teaches how you can earn more stars through particular behaviors:

Insight is no longer relevant if all that is tracked and regulated is behavior. The age of reason finally comes to an end. This, however, could profoundly transform our understanding of a democratic and free society. [...] Individual insight promised a change in behavior and thus the potential for change in society. Now it seems that people do not have to be illuminated but simply regulated by points and badges in order to make them fitter, happier, and more productive (Schrape, 2014, p. 43).

The design of technology that is intended to be comfortable and intuitive to use, and the implementation of gamification tools, suggest that the design is informed by the aim to incorporate or automate human behavior. Developers even explicitly draw on the psychological theory of operant conditioning for the construction of their products, as Schüll (2014, pp. 147–154) pointed out in her study on slot machines. Those who have worked for Silicon Valley companies, such as Tristan Harris, who worked for Google, have spoken out about how tech companies have exactly this targeting of automated behavior in mind.<sup>1</sup> The user of such technologies "delves into the physical unconscious through processes of habitualization and normalization" (Kaerlein, 2018, p. 263, translated by DK). In that framing, the learner and their behavior are understood as a trivial machine working by reflex

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<sup>1</sup> <https://www.humanetech.com/who-we-are#our-story>

mechanisms (Meyer-Drawe, 2005, p. 51), whereby algorithmic feedback loops can be conceptualized as learning processes.

This is not to suggest that these design elements within educational technologies are always intended to turn humans into – in exaggerated terms – Pavlovian dogs, as we will show in the second part. Nevertheless, the original idea of implementing gamification and optimizing tools rose from a hope within the industry to significantly influence and predict human behavior, to increase data and thereby financial gains by keeping users logged into platforms. We will outline below what this means for educational settings.

## 14.4 Algorithmic Systems and “Bildung”

Algorithmic systems have changed the relationship between knowledge, learning, and education in the narrower sense of “Bildung”. Nordenbo characterizes Bildung as “the standard German understanding of the concept as an educational idea, a person has acquired Bildung only if he or she has assisted actively in its formation or development. In other words, in the educational context, the concept of Bildung contains a reference to an active core in the person who is gebildet” (Nordenbo, 2002, S. 341). In opposition to this idea of individual educational development, technical systems address all human beings throughout all generations and lifespans – from school children to adults and into old age – in the same way. There is no room to address the particular pedagogical values and needs of different stages of development, life courses and biographies. By implementing digital software in the education process, former routines are challenged and changed. In most cases, they accelerate a process that characterizes the general school system: The need to equip students with the necessary certificates for the labor market often narrows the time available to strengthen a pedagogical process of understanding. It is not uncommon to rush through the curriculum in view of the assessment of performance. Digital systems of education intensify this process in ways that have an impact on the relation between teacher and students. It is no longer the human being who is responsible for guiding the teaching process, but a technical system that, in general, cannot be adapted by the teacher.

Most algorithmic systems, for example, work with temporal immediacy; the systems answer immediately, but also provoke immediate user reactions. This stands in contrast to our understanding of learning and “Bildung”, a process that takes time. The systems require children and adolescents to react as fast as the digital system does and thereby give the students less time to reflect on the questions or tasks presented by the software. In the following, we will exemplify how these problematic aspects and structures of algorithmic logics are implemented and designed within learning tools, and how they move educational routines away from learning and “Bildung”. From a long-term perspective, digital systems are, more than ever, pushing the next generation in a direction of becoming functional, compliant participants



and users of the algorithmic systems designed by and for the GEI. The increase in the use of such systems in schools means an increased adaptation of students to such systems and the ways of learning they require. Students are trained to react as fast as possible by being asked for immediate answers. For such systems, thinking is just an interfering variable because it takes time. For an education process guided by pedagogy, however, thinking is the central action that should be cultivated in school. When focusing on the school context, and more precisely on teaching and instruction processes on the one hand and on learning processes on the other hand, it is necessary to step back and elaborate the theoretical assumptions made about these processes. Such assumptions will help us to identify where and how digital tools and their algorithmic structures intervene and modify teaching and learning processes in classrooms. By doing so, the concept of “Bildung” – a widespread term in the German (and European) discourse that has been in use for 200 years – will serve as a tool to criticize the limitations and restrictions of algorithmically shaped learning platforms.

## 14.5 A Theory of Teaching as Critique of Digital Learning Tools

In the German discourse on learning, the theory of teaching plays a central role that has become increasingly recognized in the European and international context (Uljen & Ylimaki, 2017). Teaching is understood as a complex system of actions that consists of “Bildung”, education (“Erziehung”) and didactics (Gruschka, 2009, 2013, 2019). All three aspects are interwoven in the action of teaching. Gruschka calls it “an integrated case of constellation” (Gruschka, 2019, p. 680).<sup>2</sup> “Bildung” is the reason for and target of teaching. It “deals with all the problems of understanding which emerge from the content of the lesson” (Gruschka, 2019, p. 675). Education is not only necessary for the overall development of the child and adolescent. In the school context, the teacher’s task is to help students to orient themselves towards the target of “Bildung”. Didactics refers to the way in which the teacher arranges both the target of “Bildung” in terms of the school topic or school subject and the educating aspect that helps the student to focus. Didactics includes decisions about “which part of the knowledge has to be presented” at what point in time and to what amount (Gruschka, 2019, p. 676).

“Bildung” is the central category in the theory of teaching (Horlacher, 2004), but it does not refer only to the product of a successful teaching and learning process. “Bildung” differs from formation or training and is still tied to the classical 19<sup>th</sup>

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<sup>2</sup>Gruschka refers to Theodor W. Adorno’s term of “constellation” as a theoretical concept to highlight the interwoven structure of the term and its non-identical, cf. Adorno, 1997.



century definition offered by Wilhelm von Humboldt.<sup>3</sup> Humboldt defined “Bildung” as the “most independent interaction (interdependency) between the individual and the world” (Humboldt, 1956/1793, p. 29; translated by SJ). This interplay involves how one deals with the world and its objects as well as how one expresses oneself in, constructs and forms the world. In this sense, “Bildung” refers to the possibility as much as the need of the individual to understand and to put the necessary effort into the process of understanding. This relation is a fundamental process of self-formation that needs to be guided and supervised by teachers as we hold onto the idea of effectiveness to instruct or train groups of students by one teacher. “Bildung” is seen as any act or experience that has a formative effect on the mind, character, or physical ability of an individual. Given a close relation to the school context, it is an instrument by which individuals become autonomous, emancipated from societal – and historically developed – structures by designing and refiguring them. The individual learns to understand these structures and his or her own involvement in them. School is linked to this task by supporting the student to become familiar with and to question and reflect on them. In the long term, this process should lead to the possibility of developing society as a whole and changing society for the good of everyone. From this perspective, “Bildung” is understood as an “endless task of developing, unfolding, and enlightening the human mind and making real the independence of human will and action from natural and social determinations, coercion, and constraints” (Masschelein & Ricken, 2010, p. 127). School is therefore the central institution of a society and a nation state for the development of the individual, society, and nation. “Bildung” is not oriented to the student as a learner only, but to the student as a human being and future citizen (cf. Jornitz & Timm, 2021). Hence, the “concept of Bildung always contains an understanding of what is human (an understanding of humanity) and how we can attain this. It always implies an idea of humanity and a knowledge and representation of that which is unsatisfactory or insufficient” (Masschelein & Ricken, 2010, p. 131).

In this respect, the critical discourse on data in education must be extended to the core of education itself: to “Bildung”, sense-making and understanding as the leading principles and targets of every education process.

## 14.6 The Misconnection of Digital Tools with “Bildung”

Given the centrality of “Bildung” for a theory of teaching, it makes sense to ask how each didactic tool contributes to this process. In the following, we will exemplify how the digital transformation of teaching materials changes not only the teaching process and the relation between teacher and students, but also in what respect the target of “Bildung” itself is changed and transformed by these algorithmic

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<sup>3</sup>For the development of the term *Bildung* in Germany and Europe and its relation to theological as much as aesthetic debates, see Horlacher, 2004 and 2011. As an example for the German discussion see Rittelmeyer, 2012.

structures. Most of the digital tools in the German and the global market are coded for specific learning tasks, e.g. for training reading or math skills (Jornitz, 2015). These tasks are grouped into series that are relevant for the content of a specific school subject. Students have to click on a task, choose the right answer from a number of given answers or type the results into a defined space. Upon pressing a button, the software tool decides immediately if the answer is right or wrong. The next step depends on the structure of the program. Ultimately, the idea underlying most of the learning programs is to guide students through a one-way-street from one task to another. Every series of tasks ends with a summarized evaluation of the students' answers, before the next series appears on the screen. These tasks could mean another series of the same, or different ones at a higher or lower level depending on the earlier performance. In a coded learning tool of sufficient complexity, this structure of tasks and evaluated answers could be endless.

In the structure described above, the idea of a learning process is pre-designed by the developers of the learning tool. As outlined before, such a digital structure defines the student as a simple machine that is willing to move from one task to another, motivated only by collecting the coins, stars or points rewarded by the computer system. We could say that the idea of *Bildung*, as a process that supports the student to a better understanding of a certain aspect of life, is in such digital systems reduced to work on tasks. But the delivery of such tasks can be justified only if they provide a necessary piece in the process of constructing understanding. We argue in the following that systematic elements of most of these education tools present significant obstacles to the process of *Bildung*.

To illustrate more precisely how the topic of “*Bildung*” is addressed - or better, neglected - with these learning tools, we want to focus on three aspects: 1. immediate response, 2. reward systems, and 3. performance representation via dashboards.

### ***14.6.1 Immediate Responses in Learning Software***

Learning software offers the advantage of providing an immediate response on whether a students' answer is right or wrong. Working with paper and pencil and individual exercise books, such prompt response and feedback is not possible. Often the answers are not available to the students until the teacher delivers them to the whole class. In terms of the outlined teaching theory, such a feedback process is framed by the norms of the social relations between the teacher and the class. Students have learned that they need to wait until the answers are given to the whole class and that they have to compare the given answers with their own answers in their exercise book afterwards. They may need to reflect on the answer and why it is right or wrong. This creates an in-between phase that could be pedagogically used as a time for reflection. Didactically it means that the situation of solving the given tasks and evaluating them has to be designed. The time lag between the two phases means they are disconnected and must be re-connected by the teacher's didactic

effort. At this moment, “Bildung” could happen if a student realizes something fundamental about the content involved in the tasks.

Yet, most teachers and students experience this time lag as unsatisfying. The student wishes to know if their solution is right or wrong and continue with the next task. Teachers cannot be aware of all individual mistakes and have to focus on explaining the correct solution. In most lessons, the time lag is created as a pause, rather than the chance to think about the content of the task. It is easy for the digital tool to fill the gap, therefore, with its algorithmically given immediate feedback. The construction of the evaluation process is pedagogically different.

To give a simple example, we refer to a famous reading comprehension platform in Germany called Antolin (Jornitz & Leser, 2018). The core idea is to distribute multiple-choice questions about a book to students. When an answer is chosen, the feedback is given in the following way (Fig. 14.1).

The bottom line of the screen in green indicates a correct answer; if the answer is incorrect this element is colored red. We also see a tick and the German word for “correct” (“RICHTIG”), i.e. two more elements that confirm the correctness of the answer. Consequently, the student is rewarded six points (“6 Punkte”). The design of this reading comprehension software focuses attention on the bottom line but not necessarily on the answer itself. The design does not reveal why the student chose the correct or incorrect answer, or why and in what sense the answer is correct. The content of the book, the storyline itself and, moreover, the reading competencies become a marginal focus. The didactic form of the quiz in the digital version focuses on the right and wrong only – no matter what it means. In the example, the framing of fictional content as being right and wrong gives the impression that it is a hard fact to learn that “Niklas is the most successful girls’ enchanter in Scandinavia”



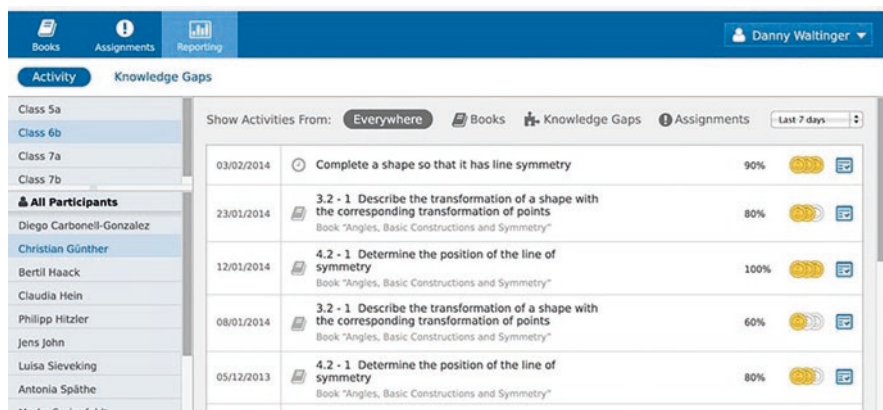
**Fig. 14.1** Quiz on the book “Niklas liebt doch nicht jeden”. (Source: Test quiz taken from the Antolin website: <https://antolin.westermann.de/>)

(and not a “bogey thrower” or “school newspaper editor”). The fact that the fictional content is a means to train reading competencies becomes obsolete. The immediacy of the answer urges the student to take the evaluation as a given fact and click to the next task. The program invokes a speed that entices the students to fast decision making but hinders proper reflection on the task and its question. Software tools that are coded in the manner of this example reduce the possibility of “Bildung” to the evaluation feedback itself. Students are trained to make decisions rapidly without reflecting on right or wrong answers. Didactically speaking, the content of the book becomes incidental, and therefore the room for “Bildung” in relation to the piece of literature diminishes.

### 14.6.2 Reward Systems as Replacements of (Pedagogical) Grading

Feedback is an essential aspect of learning software, but it is obviously not sufficient in itself. Most of the products accredit results in terms of points, coins or stars that create a specific reward system. Whereas in the school context grades and marks are the typical currency, coins and points are more akin to the gaming industry and its products. Students’ leisure time experience with such reward systems might be a motivation for the software engineers of learning tools to use these familiar elements. However, while such a system might be helpful in leisure activities, in the school context it competes with the topic of the lesson or the topic of the digital software sequence.

A screenshot of the German product for mathematics learning, *bettermarks*, illustrates an example that is also developed for the international market (Fig. 14.2).



The screenshot shows the 'bettermarks' interface. At the top, there are tabs for 'Books', 'Assignments', and 'Reporting'. The user 'Danny Waltinger' is logged in. Below the tabs, there are buttons for 'Activity' and 'Knowledge Gaps'. On the left, a sidebar lists 'Class 5a', 'Class 5b', 'Class 7a', 'Class 7b', and 'All Participants'. Under 'All Participants', several names are listed: Diego Carbonell-Gonzalez, Christian Günther, Bertil Haack, Claudia Hein, Philipp Hitzler, Jens John, Luisa Sieveking, and Antonia Späthe. The main area shows a table of activities with columns for date, description, and progress. The activities are filtered by 'Everywhere' and 'Last 7 days'.

| Date       | Activity Description                                                                                                                              | Progress |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 03/02/2014 | Complete a shape so that it has line symmetry                                                                                                     | 90%      |
| 23/01/2014 | 3.2 - 1 Describe the transformation of a shape with the corresponding transformation of points<br>Book "Angles, Basic Constructions and Symmetry" | 80%      |
| 12/01/2014 | 4.2 - 1 Determine the position of the line of symmetry<br>Book "Angles, Basic Constructions and Symmetry"                                         | 100%     |
| 08/01/2014 | 3.2 - 1 Describe the transformation of a shape with the corresponding transformation of points<br>Book "Angles, Basic Constructions and Symmetry" | 60%      |
| 05/12/2013 | 4.2 - 1 Determine the position of the line of symmetry<br>Book "Angles, Basic Constructions and Symmetry"                                         | 80%      |

**Fig. 14.2** Overview of activities in *bettermarks*; screenshot from the website. (Source: Website of *bettermarks*: <https://bettermarks.com/how-it-works/> [November 2020])

The overview of activities (see Fig. 14.2) is not only listed by topics and dates, but also shows coins and stars achieved in the column on the right, which symbolizes how successfully the student has performed on his or her tasks. The yellow color and the size of the symbols draw attention more readily than the percentage figure given. The percentage on the right, however, is another new metric that enters the classrooms and is the visual expression of quantification processes that characterize algorithmic systems, as outlined above. Sometimes this list is complemented by points, which are more in line with common teacher feedback, but often the learning software pre-defines the number of points available for a certain task and cannot be modified by the teacher. The whole system of evaluation, feedback and reward, therefore, is taken over by the software. A new “currency” is thereby introduced to the school context, one that is structured according to software games, not as pedagogical feedback from which to analyze mistakes, help to overcome misunderstanding or strengthen the knowledge and understanding of the content or subject. As these metrics are used by every digital tool or website, how we teach will be changed fundamentally. Holding onto the established form of school grading is nearly impossible if the core of instruction is replaced by solving tasks that are distributed by digital learning platforms. This metric will at the same time change the object or content itself. Its value will be quantified and measured but not situated in relation to understanding its importance for life or the development of society.

From the perspective of teaching theory, such a reward system eliminates the content and ties the student’s self-understanding as much as possible to the system and the distributed metrical data. The inner logic of the software needs a student who wants to get as many coins and stars as possible: within games that is part of the fun, but within the school context it negates the aspect of “Bildung”, because the reason why the student got his or her reward is marginalized. The reward symbols mask what topics the student has understood or what abilities or competences he or she has acquired. The student is constructed as a simple machine dependent on the stimulus to show a certain response. Education is limited to the motivation to focus on the next series of tasks the software provides. Consequently, the inner logic of a specific aspect of learning content disappears. It is deconstructed into single tasks that are then hard to reassemble in connection to each other again.

### ***14.6.3 Performance Representation via Dashboards***

Most digital learning platforms or instruments present their data via a so-called dashboard, an easy to understand visualization of key data. Rather than users running their own calculations, the task is given to the algorithm underlying the data mining systems (Williamson, 2016). Therefore, it is not the user of the software system who determines the algorithm by which data are processed and calculated. Teachers, students and parents can ultimately only work via the dashboard presentations. In most cases, attention is guided via recognizable color cues ranging, e.g. red = worst, green = best. A study about health measurements showed that participants



**Fig. 14.3** Presentation of the Quiz result. (Source: Test quiz taken from the Antolin website: <https://antolin.westermann.de/>)

attached a far greater importance to the visualizations than to the oral contextualization of a measurement expert: their measurement output visualized in red (within green to yellow to red colour schemes) caused some of them weeks of distress, in spite of the framing and relativization of the results by the measurement expert (Klinge et al., 2020).

Decision-making processes are thus pre-empted by the symbolism of the color. A simple example is taken again from the German reading competence platform, Antolin (Fig. 14.3).

The visual presentation of the quiz result (see Fig. 14.3) shows colors, a diagram, the number of right and wrong answers and the percentage of a so-called “performance” (in German: “Leistung”). Without delving deeply into an analysis of the result itself, the presentation is constructed as intuitively understandable. The red elements in this example are bigger than the green ones, the red numbers are higher than the green ones – that is enough for a rapid evaluation of the student’s performance. Because of Antolin’s simple structure the numbers can be easily connected to the number of quiz questions, but the percentage measurement of achievement doesn’t directly correlate with it. While the percentage itself is compared to 100% and is easily interpretable, i.e. a low number reflects low performance, how the percentage has been calculated is not clear. The visual presentation of the performance conceals the underlying logic. Students as well as teachers become conditioned to such performance visualization. Regardless of the metrics or the means of calculation, the message is that performance should be 100% in a totally green



circle.<sup>4</sup> In such diagrams, the content underlying “Bildung” disappears again; the software instruments ideally prepare the user for any task that appears next on the screen. Children are not prepared to query, but to respond.

## 14.7 The Trias of Sense Making, “Bildung” and Education Technology Within a Geopolitics of Knowledge

Digital technologies are not neutral: the programming of algorithms and the design of computer technology always reflect models of the social world, certain practices and anthropological ideas (Jornitz & Engel, 2021; Wilmers & Jornitz, 2021). Discourses of quantification and optimization of education have encouraged certain technological developments and implementations of digital technologies in educational settings. The “imaginary” of education and data processing practices view learning “as quantifiable, measurable, actionable, and therefore optimizable” (Williamson, 2017, p. 120).

In turn, educational practices with and within digital tools will influence human meaning construction and social structures. The expectations of instant feedback and rewards for right answers provided by the real-time operability of algorithmic data processing will accompany how students learn content and how they learn what education is (and how they are taught). Gamification tools will likewise change the framing of education and competition and determine the educational potential of *games* as spheres of pretending and trying out different scenarios.

The use of digital instruments in schools is often linked to a hope for more individualized teaching practices. Selwyn points out that “digital technologies are seen as enhancing student’s control over the nature and form of what they do, as well as where, when and how they do it” (Selwyn, 2011, p. 16). In consequence a teacher’s role would mean assisting the student in task management. Such digital learning systems also reveal that they model and process school curriculum topics in rather simple ways. Students are trained by multiple-choice formats rather than being educated in critical reflection and understanding of reasoning (Rittelmeyer, 2018). Moreover, digital instruments seldom serve to personalize and individualize in an authentic manner. Students are grouped according to their level of knowledge (low, medium, high) and provided with respective tasks from an integrated database of the platform or app. It is an open question whether this procedure makes students remain at a certain level of knowledge or if such a tool assists them in reaching a higher level. It has been argued that the use of digital technology within learning

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<sup>4</sup>Figure 14.3 shows that the two metrics do not correspond; they are contradictory and mathematically at least irritating: while the number of points is negative (here: minus 42), the measured performance (“Leistung”) is stated as 26% – but it is mathematically not possible to get a positive result based on negative numbers. The two systems, to add negative points for wrong answers on the one hand, and to add the (positive) points for right answers and transfer them into a performance-percentage on the other, do not build a coherent system.



processes with their characteristics of quantification, simplification and stimulus-response design do not facilitate individual development.

The idea of “Bildung” entails that learning processes need time and room for ambiguity. Education in this sense is not only about knowledge transfer but also oriented to the student as a human being and future citizen and also to questions about the good life. If we follow Masschelein and Ricken (2010), we understand “Bildung” as “(1) the formation of the self in relation to itself [...]; (2) the formation of the social, that is, as a particular formation of the self in relation to others [...]; (3) being in relation to the world” (p. 130). Against this background, digital tools, first, frame the relation to oneself in terms of ‘earned’ points and coins. The personal achievement in school is evaluated by an algorithmic-based machine rather than by a responsible teacher. Second, the self in relation to others is constructed via competition within an abstract reward system and without communication ‘between the lines’. This competition is not linked to understanding or sense-making of certain subjects, but to the pure contrast of one against the other. And third, by adapting students to such systems, it seems that the world in this teaching setting gives immediate answers about right and wrong in color schemes of green and red. In such a setting, the truth is always immediately at hand, and in the hands of those who design and code the programmes.

Such a technological system for learning can be distributed throughout the world, no matter of country or school culture. These digital technologies are built around a global economic concept and transport a fusion of market-logics, algorithmic functionality and Ed-Tech narratives that promote the idea that learning is actively driven by technology (Mertala, 2020). The emergence of a global education industry and for-profit actors promoting these tools within the educational sector not only change students’ perspectives on the education and teaching process but also that of teachers and schools. These kinds of market-driven technical developments will shape our picture of future schooling fundamentally. We do have different school systems and different kinds of learning materials. Widespread standardization of practices – like the OECD’s programme “PISA for Schools” – lead to worldwide processes in which students will learn with the same materials and programmes. The didactical procedures as much as the types of tasks presented in digital learning platforms as part of a digital geopolitics of knowledge will deny differences in cultures and curricula. Digital technologies have the power to streamline pedagogical settings worldwide.

While this offers a chance to enable communication and build networks between actors in different educational systems, as social relations are not defined through belonging (Hall, 2017) but through integration and disintegration within a network (Wittel, 2001, p. 51), programs such as the OECD’s Programme for International Student Assessment (PISA) cannot rely on the national curricula for its items so far, but if certain educational technology tools are implemented in the classrooms, they standardize what and how it is taught worldwide – and makes assessments easier in the future to measure student’s performances in the classrooms.

As researchers we do have a duty to think about such a global alignment of educational teaching processes.

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